ASSIGNMENT 4: WATERMARK FORGERY ATTACK

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Github code: https://github.com/arcnova2095/wm$_{forge}$

1 ASSIGNMENT

This assignment considers forging invisible watermarks under black-box conditions. For each of the eight watermarking schemes (`WM_1`–`WM_8`), 25 watermarked images carrying the same hidden message and 200 clean images are provided. The goal is to transfer the watermark onto batches of 25 clean images while preserving visual quality. Performance is measured by the final score, $S_{\text{final}} = S_{\text{det}} \times S_{\text{qlt}}$, where S_{det} evaluates watermark recovery accuracy and $S_{\text{qlt}} = e^{-8 \cdot \text{LPIPS}}$ penalizes perceptual distortion. Consequently, successful watermark forgery requires balancing robust watermark transfer with minimal visible artifacts.

2 METHODOLOGY

Scheme identification. We have tested every candidate decoder we had access to - imwatermark's dwtDct, dwtDctSvd, and rivaGan modes, three TrustMark variants (Q/P/C), and blind_watermark. We tested them at a few payload lengths, against every WM_n source folder. For each of the candidate, we decoded all the 25 images and took a majority vote per bit. We measured how often individual decodes agreed with that vote. Agreement above 85% confirmed a match, since a mismatched decoder gives close to 50% (random) agreement.

Decode-and-re-encode. For the four schemes we have identified this way (WM_1: dwtDct, WM_2: RivaGan, WM_7: TrustMark-Q, WM_8: TrustMark-P, all above 98% agreement). The attack is mechanical: decode the sources, majority-vote the bits into one clean message. And re-encode it onto the matching clean targets with the same library's encoder. No strength tuning is needed, since we have reused the scheme's own encoder and its built-in trade-off.

Blind fingerprinting. WM_3, WM_6 matched none of the decoder, so we estimated the watermark itself as a shared additive pattern across the 25 source images. Following the "simple averaging" idea of Yang et al. (NeurIPS 2024): different content but the same watermark means that the averaging residuals cancels the content and leaves the watermark. We have computed a spatial high-pass residual (image minus a blurred copy) and a frequency-domain band-pass residual (via FFT) and aggregated both across the batch with the median. Then rescaled using median/MAD statistics for robustness and clipped the combined signal before adding it to each clean target.

DESIGN CHOICES AND HYPERPARAMETERS FOR BLIND FINGERPRINTING

We switched from a mean/std aggregation to a median and MAD-based rescaling after the outlier images skewed the fingerprint and caused visible artifacts. Median/MAD are far less sensitive to a few unusual images or pixels. We have combined a spatial high-pass residual with a frequency-domain band-pass residual, weighted per scheme, since we did not know in advance whether a given scheme embeds spatially or in the frequency. For the identified schemes, we have disabled the TrustMark's built-in ECC (use_ECC=False) when re-encoding, since our majority-voted message was already denoised and the ECC framing only wasted the payload budget. For WM_3,WM_6 we swept blur scale, spatial/frequency strength weights, frequency band cutoffs, and clipping bound, picking per scheme whichever configuration maximized S_{final} (since S_{det} and S_{qlt} trade off directly). We also tried the approach from the paper WMCopier: Forging Invisible Image Watermarks on Arbitrary Images, but it was less effective. Yang et al.'s simple-averaging approach remained our best strategy for the unidentified schemes.

2.1 Summary of Results

Table **??** summarizes the performance of the evaluated watermarking schemes. Overall, the methods achieve an average S_{final} of **0.8978**. **TrustMark (Q)** (WM_7) achieves the highest score (**0.9888**), closely followed by **TrustMark (P)** (WM_8) (**0.9884**). The fingerprint-based schemes (WM_3–WM_6) achieve perfect detection ($S_{\text{det}} = 1.0000$), with performance differences arising from image quality. In contrast, *imwatermark (dwtDct)* records the lowest final score (**0.8057**). Overall, TrustMark-based methods provide the best balance between watermark detectability and visual quality.

Scheme	**Method**	S_{det}	S_{qlt}	S_{final}
WM_1	imwatermark (dwtDct)	0.8525	0.9451	0.8057
WM_2	imwatermark (RivaGan)	0.9888	0.8816	0.8717
WM_3	unknown – fingerprint	1.0000	0.8092	0.8092
WM_4	unknown – fingerprint	1.0000	0.9770	0.9770
WM_5	unknown – fingerprint	1.0000	0.9714	0.9714
WM_6	unknown – fingerprint	1.0000	0.7704	0.7704
WM_7	TrustMark (Q)	1.0000	0.9888	0.9888
WM_8	TrustMark (P)	0.9976	0.9908	0.9884

Table 1: Majority Bit Agrrement Table

3 Conclusion

The best-performing setup combined two complementary attacks: decoder-based majority-vote re-encoding for the four publicly identifiable watermarking schemes, and blind residual averaging for the four unidentified schemes. While the former could be attacked using publicly available decoders, the latter required balancing watermark removal strength against image quality. These results highlight a key limitation of invisible watermarking as a standalone provenance mechanism. Publicly available schemes are vulnerable once their decoding algorithms are known, while even unknown schemes can be weakened through statistical attacks when the same watermark is embedded across multiple images. Consequently, watermarking should be complemented with stronger provenance mechanisms, such as cryptographic signatures, active defenses, or ownership verification techniques.